

41. Which of the following elements show the given properties?

- (i) All elements are metals.
- (ii) Most of the elements form coloured ions, exhibit variable valence and paramagnetism.
- (iii) Frequently used as catalysts.

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- (a) Chalcogens
- (b) Transition elements
- (c) Inner transition elements
- (d) Representative elements

Answer: (b) Transition elements

Explanation: Transition elements are metals and commonly show variable oxidation states, coloured ions, paramagnetism and catalytic behaviour because of their partially filled d-orbitals.

42. Element X forms a chloride with the formula XCl_2 , which is a solid with a high melting point. X would most likely be in the same group of the periodic table as

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- (a) Na
- (b) Ca
- (c) Al
- (d) Si

Answer: (b) Ca

Explanation: A stable chloride XCl_2 suggests X commonly forms X^{2+} . A high-melting ionic chloride is typical of group 2 metals such as calcium.

43. Which is a metalloid?

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- (a) Pb
- (b) Sb
- (c) Bi



(d) Zn

Answer: (b) Sb

Explanation: Antimony (Sb) lies along the staircase boundary between metals and non-metals and is classified as a metalloid. Pb, Bi and Zn are metals.

44. Which of the following elements are found in pitch blende?

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- (a) Actinium and protoactinium
- (b) Neptunium and plutonium
- (c) Actinium only
- (d) Both (a) and (b)

Answer: (a) Actinium and protoactinium

Explanation: Actinium and protactinium occur naturally in uranium ores such as pitchblende as members of radioactive decay series. Neptunium and plutonium are primarily regarded as synthetic/transuranium elements in this syllabus context.

45. The order of increasing sizes of atomic radii among the elements O, S, Se and As is

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- (a) $\text{As} < \text{S} < \text{O} < \text{Se}$
- (b) $\text{Se} < \text{S} < \text{As} < \text{O}$
- (c) $\text{O} < \text{S} < \text{As} < \text{Se}$
- (d) $\text{O} < \text{S} < \text{Se} < \text{As}$

Answer: (d) $\text{O} < \text{S} < \text{Se} < \text{As}$

Explanation: Atomic size increases down group 16: $\text{O} < \text{S} < \text{Se}$. In period 4, As lies to the left of Se, so As has the larger radius. Thus $\text{O} < \text{S} < \text{Se} < \text{As}$.

46. The van der Waals and covalent radii of fluorine atom respectively, from the following figure, are

- (a) 219 pm, 72 pm
- (b) 75 pm, 72 pm
- (c) 147 pm, 72 pm



(d) 147 pm, 144 pm

Answer: (c) 147 pm, 72 pm

Explanation: For fluorine, the covalent radius is half the F–F bond distance and is about 72 pm. The non-bonded van der Waals radius is much larger, about 147 pm.

47. As we move across the second period from C to F, ionisation enthalpy increases, but the trend from C to F for ionisation enthalpy is $C < O < N < F$. This is because

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- (a) the atomic radius of O is greater than the atomic radius of N
- (b) the electronic configuration of N is more stable than the electronic configuration of O
- (c) the atomic radius of N is greater than the atomic radius of O
- (d) none of these

Answer: (b) the electronic configuration of N is more stable than the electronic configuration of O

Explanation: Nitrogen has the especially stable half-filled $2p^3$ configuration. Oxygen is $2p^4$, so one p-orbital contains a paired set of electrons; repulsion makes removal of one electron easier, giving O a lower ionisation enthalpy than N.

48. Covalent radii of atoms vary in the range of 72 pm to 133 pm from F to I, while those of noble gases from He to Xe vary from 120 pm to 220 pm. This is because, in the case of noble gases

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- (a) covalent radius is very large
- (b) van der Waals radius is considered
- (c) metallic radius is considered
- (d) none of these

Answer: (b) van der Waals radius is considered

Explanation: Noble gases normally do not form covalent bonds with one another. Their atomic size is therefore estimated from non-bonded contact distances, i.e. van der Waals radii, which are larger than covalent radii.



49. Why is the size of an anion larger than that of the parent atom?

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- (a) Due to increased repulsion among the electrons.
- (b) Due to decrease in effective nuclear charge.
- (c) Due to increase in effective nuclear charge.
- (d) Both (a) and (b)

Answer: (d) Both (a) and (b)

Explanation: When an atom gains electron(s), electron–electron repulsion increases while the same nuclear charge must attract more electrons. The effective attraction per electron decreases, so the anion expands and becomes larger than the parent atom.

50. In the ions P^{3-} , S^{2-} and Cl^{-} , the increasing order of size is

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- (a) $Cl^{-} < S^{2-} < P^{3-}$
- (b) $P^{3-} < S^{2-} < Cl^{-}$
- (c) $S^{2-} < Cl^{-} < P^{3-}$
- (d) $S^{2-} < P^{3-} < Cl^{-}$

Answer: (a) $Cl^{-} < S^{2-} < P^{3-}$

Explanation: P^{3-} , S^{2-} and Cl^{-} are isoelectronic, each with 18 electrons. In an isoelectronic series, radius decreases as nuclear charge increases; therefore Cl^{-} is smallest and P^{3-} is largest.

51. Arrange the following in increasing order of ionic radii: C^{4-} , N^{3-} , O^{2-} and F^{-} .

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- (a) $C^{4-} < N^{3-} < O^{2-} < F^{-}$
- (b) $N^{3-} < C^{4-} < O^{2-} < F^{-}$
- (c) $F^{-} < O^{2-} < N^{3-} < C^{4-}$
- (d) $O^{2-} < F^{-} < N^{3-} < C^{4-}$



Answer: (c) $F^- < O^{2-} < N^{3-} < C^{4-}$

Explanation: All four ions have 10 electrons. Greater nuclear charge pulls the same number of electrons more strongly, so F^- is smallest and C^{4-} is largest.

52. The correct order of melting point is

- (a) $Be > Mg > Ca > Sr$
- (b) $Sr > Ca > Mg > Be$
- (c) $Be > Ca > Mg > Sr$
- (d) $Be > Ca > Sr > Mg$

Answer: (d) $Be > Ca > Sr > Mg$

Explanation: The approximate melting points are $Be \approx 1287^\circ C$, $Ca \approx 842^\circ C$, $Sr \approx 777^\circ C$ and $Mg \approx 650^\circ C$. Hence the decreasing order is $Be > Ca > Sr > Mg$.

53. The correct decreasing order for metallic character is

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- (a) $Na > Mg > Be > Si > P$
- (b) $P > Si > Be > Mg > Na$
- (c) $Si > P > Be > Na > Mg$
- (d) $Be > Na > Mg > Si > P$

Answer: (a) $Na > Mg > Be > Si > P$

Explanation: Metallic character increases down a group and decreases from left to right across a period. Na is therefore most metallic among these, followed by Mg; Be is more metallic than the p-block elements Si and P, while P is the least metallic.

54. Arrange S, P and As in order of increasing ionisation energy.

- (a) $S < P < As$
- (b) $P < S < As$
- (c) $As < S < P$
- (d) $As < P < S$

Answer: (c) $As < S < P$

Explanation: First ionisation enthalpy generally decreases down a group, so As is lower



than P. Within period 3, P has the stable half-filled $3p^3$ configuration, making its ionisation enthalpy slightly higher than that of S ($3p^4$). Thus $As < S < P$.

55. Of the given electronic configurations for the elements, which electronic configuration indicates that there will be an abnormally high difference in the second and third ionisation energies for the element?

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- (a) $1s^2 2s^2 2p^6 3s^2$
- (b) $1s^2 2s^2 2p^6 3s^1$
- (c) $1s^2 2s^2 2p^6 3s^2 3p^1$
- (d) $1s^2 2s^2 2p^6 3s^2 3p^2$

Answer: (a) $1s^2 2s^2 2p^6 3s^2$

Explanation: This is Mg. After two electrons are removed, Mg^{2+} has the stable noble-gas configuration [Ne]; removing the third electron would break this closed shell, causing a very large jump between the second and third ionisation energies.

56. Which of the following statement(s) is/are incorrect?

- (i) Ionisation enthalpy is expressed in units of kJ mol^{-1} .
- (ii) Ionisation enthalpy is always positive.
- (iii) Second ionisation enthalpy will be higher than the third ionisation enthalpy.

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- (a) Only (ii)
- (b) Only (iii)
- (c) (ii) and (iii)
- (d) None of these

Answer: (b) Only (iii)

Explanation: Ionisation enthalpy is expressed in kJ mol^{-1} and is positive because energy must be supplied to remove an electron. Successive ionisation enthalpies increase, so the third is higher than the second; statement (iii) is therefore incorrect.

57. The first ($\Delta_i H_1$) and second ($\Delta_i H_2$) ionisation enthalpies (in kJ mol^{-1}) and the electron gain enthalpy ($\Delta_{\text{eg}} H$) (in kJ mol^{-1}) of the elements I, II, III, IV and V are given below.



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The most reactive metal and the least reactive non-metal of these are respectively

- (a) I and V
- (b) V and II
- (c) II and V
- (d) IV and V

Given data:

Element	$\Delta_i H_1$ (kJ mol ⁻¹)	$\Delta_i H_2$ (kJ mol ⁻¹)	$\Delta_{eg} H$ (kJ mol ⁻¹)
I	520	7300	-60
II	419	3051	-48
III	1681	3374	-328
IV	1008	1846	-295
V	2372	5251	+48

Answer: (c) II and V

Explanation: Element II has the lowest first ionisation enthalpy (419 kJ mol⁻¹), characteristic of a very reactive alkali metal, so it is the most reactive metal. Element V has extremely high ionisation enthalpy and a positive electron-gain enthalpy, characteristic of a noble gas; it is therefore the least reactive non-metal.

58. Among the following transition elements, pick out the element/elements with the highest second ionisation energy.

- (A) V (At. no. 23)
- (B) Cr (At. no. 24)
- (C) Mn (At. no. 25)
- (D) Cu (At. no. 29)
- (E) Zn (At. no. 30)

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- (a) (A) and (C)
- (b) (B) and (D)
- (c) B and E
- (d) Only (D)



Answer: (b) (B) and (D)

Explanation: Cr loses its 4s electron first to form Cr^+ with the stable $3d^5$ configuration, so the second ionisation disrupts a half-filled d-subshell. Cu^+ has the stable $3d^{10}$ configuration after first ionisation, so its second ionisation also requires unusually high energy.

59. Halogens and chalcogens have highly electron-gain enthalpy values that are generally

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- (a) negative
- (b) positive
- (c) zero
- (d) infinity

Answer: (a) negative

Explanation: Halogens and chalcogens readily accept electrons. Electron addition is usually exothermic for them, so their electron-gain enthalpy values are generally negative.

60. The correct order of electron gain enthalpies of Cl, F, Te and Po is

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- (a) $\text{F} < \text{Cl} < \text{Te} < \text{Po}$
- (b) $\text{Po} < \text{Te} < \text{F} < \text{Cl}$
- (c) $\text{Te} < \text{Po} < \text{Cl} < \text{F}$
- (d) $\text{Cl} < \text{F} < \text{Te} < \text{Po}$

Answer: (b) $\text{Po} < \text{Te} < \text{F} < \text{Cl}$

Explanation: In terms of the magnitude of electron affinity/electron-gain tendency, Cl is highest; F is slightly lower because its compact 2p shell causes greater electron–electron repulsion. Down group 16 the tendency decreases, giving $\text{Te} > \text{Po}$. Thus $\text{Po} < \text{Te} < \text{F} < \text{Cl}$.

